POTPRESS · STEM CHECK

# PotPress STEM check

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## DOCUMENT CONTROL

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|---|---|---|---|
| Document ID | PPR-STM-001 | Version | 0.4 |
| Status | Draft | Date | 2026-10-25 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/potpress | | |
| Source | docs/01b-stem-check.md @ b112366 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.4 | 2026-10-25 | Amish Chadha | D9 decided by Amish (2026-10-25): S5 and S8 at the 5.4 t working force limit (S8 148 N with 85 % efficiency, still Plausible, to confirm by EN 1005-3); S4 at the 10 t structural check case; verdict still open |
| 0.3 | 2026-10-24 | Amish Chadha | Kit 1.9.1 upgrade: eight rows for options B and C checked on their detailed models; independent review of the headline results: S5, S7 restated, S11 added (joint preload), S8 to Plausible (D9 to Amish), option B pins 40 mm; verdict open |
| 0.2 | 2026-10-23 | Amish Chadha | Four failed details fixed in the design under Amish's standing instruction of 2026-10-23 (PPR-DDR-006); fixes reviewed by an independent agent and corrected; check rerun: pass |
| 0.1 | 2026-10-23 | Amish Chadha | First STEM check (kit 1.8 upgrade), with an independent reviewer pass |

# PotPress STEM check

**Verdict: open (one row to confirm).** PotPress's working principle, a hand-pumped hydraulic press forming clay between cast molds, sits within known science, and 36 of its 37 claims are within known science (29 for option A, eight for options B and C added on 2026-10-24 from their detailed kit 1.9 models). One row is open: the independent review of 2026-10-24 showed that the pump handle force at the 6.4 t working force limit (S8) had no margin once the jack's pump and seal efficiency is counted. Amish decided D9 on 2026-10-25 ("i concur with your recommendations"): the working force limit is now 5.4 t, where the handle force is about 148 N with 85 % efficiency, and 10 t is a structural check case only. S8 stays **Plausible, to confirm** until the EN 1005-3 assessment with the bought jack's data is done. `project.yaml` records `stem_check: open` until it closes. The first check (v0.1) found four details that failed with a fix available: clay shrinkage, heavy lifts, pump handle force and the scrap alloy. Under Amish's standing instruction of 2026-10-23 ("whenever a STEM check fails propose and implement the best fix to the design and constructability"), each fix is now in the model, drawings, calculations, BOM and requirements (PPR-DDR-006). An independent reviewer then tried to refute the fixes and two were corrected, as described below. Method: STANDARDS section 20, fix rule. Calculations: `docs/04-calcs/stem_check.py`, results in `docs/04-calcs/stem_results.csv`.

## Claims checked

Table 1. Claims, governing science and verdicts.

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| S1 | Top and base beams (2 x UPN 160, S275) stay below yield at 294 kN | Physics | Strength of materials | Beam bending; S275 yield 275 MPa (EN 10025-2) | 190 MPa <= 275 MPa | **Within known science** | PPR-CAL-001 section 4 |
| S2 | 60 mm 42CrMo4 load pin carries 294 kN in bending without yield | Physics | Strength of materials | Pin in double shear and bending; 42CrMo4 quenched and tempered yield about 650 MPa | 272 MPa <= 650 MPa | **Within known science** | PPR-CAL-001 section 4 |
| S3 | M16 10.9 joint bolts carry their share of 294 kN in double shear | Physics | Strength of materials | Bolt shear resistance 0.5 x fub / gamma_M2 = 0.5 x 1000 / 1.25 = 400 MPa (EN 1993-1-8) | 117 MPa <= 400 MPa | **Within known science** | PPR-CAL-001 section 4 |
| S4 | Deflection between the molds below 1 mm at the 10 t structural check case | Physics | Mechanics and statics | Beam and column elastic deflection, E = 200 GPa | 0.66 mm <= 1 mm | **Within known science** | PPR-CAL-001 section 4 |
| S5 | A pot filter can be formed within the 5.4 t working force limit | Physics | Mechanics and statics | A low-cost press forms round-bottom pot filters with a 2 t jack (Henry, Maley and Mehta 2013); factories press flat-bottomed 280 mm pots on 20 t jack presses without a published working force. The pressed pot here is about 18 % larger than the factory form, so the force it needs is found | 2 t <= 5.4 t | **Within known science** | Pressing trials with a partner factory, starting near 2 t (decided 2026-10-02); working force limit 5.4 t (D9, decided by Amish 2026-10-25) |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | | | | in trials (decided 2026-10-02); 10 t is a structural check case, not a working force (D9, decided by Amish 2026-10-25) | | | |
| S6 | Mean pressure on the clay at 10 t is about 0.9 MPa | Physics | Mechanics and statics | Pressure = force / projected area | claimed 0.89 MPa, estimate 0.888 MPa (ratio 1.00) | **Within known science** | PPR-CAL-001 section 3 |
| S7 | One operator can pump each pressing by hand (power, at 10 t with 85 % jack efficiency) | Physics | Fluid mechanics | Work = force x distance: the load builds over the last 20 mm (half of 10 t on average) after a 90 mm approach at 0.5 t, about 1 min of pumping, 85 % pump and seal efficiency; sustained human output about 75 W | 27.8817 W <= 75 W | **Within known science** | PPR-CAL-001 section 6 (cycle breakdown) |
| S8 | Pump handle force stays within repetitive-work guidance at the working force limit of 5.4 t | Biology | Ergonomics | Handle force = jack force x piston-to-ram area ratio / lever ratio / efficiency: about 126 N at 5.4 t frictionless and about 148 N with 85 % pump and seal efficiency, inside the 150 N screening limit (the earlier 6.4 t limit gave about 175 N with efficiency); about 70 strokes a | 126 N frictionless, 148 N at 85 % efficiency, against a 150 N screen; working force limit lowered from 6.4 to 5.4 t (D9, decided by Amish 2026-10-25) | **Plausible, to confirm at TRL 3** | To confirm: EN 1005-3 assessment with the bought jack's piston, lever and efficiency data (PPR-CAL-001, next session); the two-speed pump (D6) stays open if trials need more than 5.4 t |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | | | | pressing at about 50 a minute; EN 1005-3 frequency and duration multipliers not yet applied | | | |
| S9 | Pin, beams and cast molds survive 20,000 pressings a year at 10 t | Physics | Strength of materials | Fatigue: beam welded detail category 71 MPa at 2 million cycles (EN 1993-1-9); 42CrMo4 endurance about 400 MPa; cast Al-Si about 50 MPa | 63.3333 MPa <= 71 MPa | **Within known science** | This file (stress at 10 t is one third of the 294 kN value) |
| M1 | Reference filter holds 12.30 L to the brim | Mathematics | Units and dimensional analysis | Frustum volume | claimed 12.3 L, estimate 12.3 L (ratio 1.00) | **Within known science** | PPR-CAL-001 section 2 |
| M2 | 69 pots per 6 h at a 5.2 min cycle | Mathematics | Orders of magnitude | Rate = time / cycle | claimed 69 pots, estimate 69.2 pots (ratio 1.00) | **Within known science** | PPR-CAL-001 section 6 |
| M3 | 0.1 L reads as 1.69 mm on the T-gauge | Mathematics | Units and dimensional analysis | Volume = area x height (water surface about 275 mm across) | claimed 0.0591716 m2 of water surface, estimate 0.0594 m2 of water surface (ratio 1.00) | **Within known science** | PPR-CAL-001 section 9 |
| M4 | Charge of 9.59 kg of mix per pot | Mathematics | Orders of magnitude | Factory practice 8 to 9.5 kg per filter (Rayner 2009) | claimed 9.59 kg, estimate 8.75 kg (ratio 1.10) | **Within known science** | PPR-CAL-001 section 2 |
| M5 | A push of about 850 N at 1 m tips the unanchored press (centre of mass 246 | Physics | Mechanics and statics | Restoring moment = weight x lever arm | claimed 850 N m, estimate 849 N m (ratio 1.00) | **Within known science** | PPR-CAL-001 section 11 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | mm behind the front foot edge, carriage out) | | | | | | |
| C1 | Fired pot of 4.88 kg from an 8.72 kg pressed pot | Chemistry | Physical chemistry | Mass balance: water leaves, clay loses about 10 % on ignition, rice husk leaves 15 to 20 % silica ash (ash now in CAL) | claimed 4.88 kg, estimate 4.87 kg (ratio 1.00) | **Within known science** | PPR-CAL-001 v0.11 section 2 (ash added) |
| C2 | Rice husk or sawdust burns out on firing and leaves the pores that set flow | Chemistry | Materials and polymer chemistry | Combustion of cellulose well below the 800 to 900 C firing; pore size set by burnout particle size (Heijman et al. 2015) | PotPress controls wall thickness only; recipe and firing out of scope | **Within known science** | PPR-PRB-001 |
| C3 | Aluminum mold faces and polyethylene liners do not contaminate the pot | Chemistry | Inorganic | Aluminium worn off by wet clay oxidises to alumina, already a clay constituent; the real risk is lead, cadmium, copper or zinc in scrap alloy, and wheel scrap carries lead balance weights and brass valve stems | A356 wheel scrap only, sorted before melting (weights, valve stems, steel and paint removed); a chill sample of each melt checked by spark OES, or handheld XRF for lead, before pouring; lead and cadmium within the EN 601 limits (values to confirm); recorded per heat (PR-012 verification) | **Within known science** | PPR-DDR-006 (D5); independent review 2026-10-23 |
| C4 | Silver treatment is outside | Chemistry | Environmental chemistry | WHO silver chemical fact sheet (2022): | No claim made by PotPress; | **Within known science** | Out of scope (PRB) |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | PotPress; a silver-treated filter can stay under WHO's 0.1 mg/L reference value | | | provisional reference value 0.1 mg/L, no health-based guideline value | noted for the factory | | |
| C5 | Flow changes about 2.3 % per C near 25 C | Chemistry | Physical chemistry | Darcy flow scales with 1/ viscosity of water | claimed 2.3 % per C, estimate 2.33 % per C (ratio 0.99) | **Within known science** | PPR-CAL-001 section 9 |
| B1 | PotPress makes no pathogen-removal claim; locally made pot filters cut E. coli by about 99 % | Biology | Microbiology | Brown and Sobsey 2010; WHO HWT scheme 2-star target is 2 log for bacteria | The press makes consistent pots; removal must be lab-tested per batch (stated in PRC Safety) | **Within known science** | PPR-PRC-001 Safety |
| B2 | Parts over 25 kg (female mold 39.6 kg, platen 39.6 kg, top crossbeam 38.0 kg, base beam 34.2 kg, male mold 26.3 kg) are lifted safely | Biology | Human physiology and biomechanics | Team lifting: two people carry about two thirds of their combined capacity (about 31 kg at 23 kg each), lower still at 1.5 m height and for bulky loads. M12 eye bolt in steel plate or 26 mm of Al-Si thread: tens of kN to strip against about 0.2 kN per leg | Every part over 25 kg goes on a 0.5 t chain block: female mold by two M12 DIN 580 eye bolts in its steel base plate, male mold by two in its flange top, both on a two-leg sling with shackles and the eyes shimmed into the sling plane; beams and platen in a round sling. Eye bolts out and holes plugged before pressing. The 40 kg limit in PR-008 is a transport limit | **Within known science** | PPR-DDR-006 (D3); independent review 2026-10-23 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| B3 | Silica-bearing clay dust (and rice husk ash in fired trimmings) is a respiratory hazard controlled by wet work and P2 or N95 masks | Biology | Toxicology | Crystalline silica causes silicosis; wet methods and respirators are standard controls; air monitoring shows whether they work | Safety claim only | **Within known science** | PPR-PRC-001 Safety |
| B4 | Zinc fume from welding galvanised mesh is controlled by grinding the coating off and ventilation | Biology | Toxicology | Zinc oxide fume causes metal fume fever | Stated in the build plan safety notes | **Within known science** | PPR-BLD-001 Safety |
| M6 | The fired filter comes out at the 280 mm reference size | Physics | Thermodynamics | Clay shrinks on drying and firing, typically 6 to 10 % linear for earthenware filter mixes; the mold cavity is the fired size divided by (1 - shrinkage). CMWG 2011 notes grog changing fired size enough that pots no longer fit receptacles | claimed 280 mm inner rim, estimate 280 mm inner rim (ratio 1.00) | **Within known science** | PPR-DDR-006 (D2): cavity scaled for 8 % total shrinkage (CLAY_SHRINK in cad/src/model.py), to be replaced by the partner clay's test-bar value (ASTM C326 method) before patterns are printed |
| S10 | The mix fills the cavity, air escapes and the green pot releases and survives the demold tilt | Physics | Fluid mechanics | Plastic clay pressing between two-part molds is established practice in over 50 factories; vents and liners are standard | Within known science; mold vents and release confirmed in the first pressing trials (BLD first checks) | **Within known science** | PPR-BLD-001 first checks |
| S11 | | Physics | | | | | |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | The bolted upright joints do not slip in their 18 mm holes at 294 kN (preloaded M16 10.9) | | Mechanics and statics | Slip resistance n mu Fp,C with Fp,C = 0.7 fub As (EN 1993-1-8 category C practice); slip factor 0.2 for unprepared surfaces | 36.7749 kN per bolt <= 43.96 kN per bolt | **Within known science** | PPR-BLD-001 step 8 (preload torque); PPR-CAL-001 v0.12 section 4 |
| E1 | Casting patterns print on a printer with a true 250 x 250 mm bed | Engineering and technology | Manufacturing tolerances | 486 mm pattern split into quadrants; a 250 x 210 mm or 220 x 220 mm bed does not fit | 243 mm <= 250 mm | **Within known science** | PPR-CAL-001 section 8 |
| E2 | Hand lapping to printed templates reaches +/-0.3 mm on a 400 mm casting | Engineering and technology | Manufacturing tolerances | Grinding and filing to templates reach +/-0.3 mm in toolroom practice; printed PLA templates can be off by 0.2 to 0.5 % over 400 mm, which the tolerance budget does not yet include | Within known science; practical risk carried in RSK and requirement PR-002 (at risk) | **Within known science** | PPR-CAL-001 section 7; RSK R2 |
| E3 | Guard openings meet ISO 13857 Table 4 for their distance from moving parts | Engineering and technology | Standards practice | ISO 13857:2019 Table 4 | 23 openings checked from the model; Table 7 (lower limbs) and the floor gap not yet checked; signature by a competent person is a hold point | **Within known science** | PPR-CAL-001 section 15 |
| OB1 | Option B: the toggle gives about 1.4 t at 0.35 mm from straight (0.16 t at 30 | Physics | Mechanics and statics | Virtual work on the toggle with the lever rigid on the upper link: F = T / (2 L sin theta); the kit | claimed 1.4 t, estimate 1.42 t (ratio 0.98) | **Within known science** | PPR-OPT-001 v0.3 (figures restated from the detailed model) |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | mm, 7 t at 0.014 mm) | | | 1.8 estimate (2 t at 0.35 mm) was about 40 % high | | | |
| OB2 | Option B: the 80 x 40 mm lower link does not buckle at 1.5 times 10 t | Physics | Strength of materials | Euler buckling of a pinned strut, P = pi^2 E I / L^2, E = 200 GPa | 147.1 kN <= 5264 kN | **Within known science** | This file (model link 80 x 40 x 400 mm) |
| OB3 | Option B: the 40 mm clevis pins carry 1.5 times 10 t in double shear | Physics | Strength of materials | Pin shear 0.5 x fub / gamma_M2 with fub 600 MPa (EN 1993-1-8 pin practice) | 58.5291 MPa <= 240 MPa | **Within known science** | This file (model pins 40 mm, ISO 2341 B) |
| OB4 | Option B: the 40 mm clevis pins stay below yield in bending at 1.5 times 10 t | Physics | Strength of materials | Pin bending M = F/2 (a/2 + g + b/4), sigma = M / (pi d^3 / 32), yield about 640 MPa (8.8 class pin steel); the 30 mm pins first drawn reached about 510 MPa | 198.999 MPa <= 512 MPa | **Within known science** | cad/src/options_bc.py (pins 30 to 40 mm after the independent review, 2026-10-24) |
| OC1 | Option C: two people at 300 N on the 450 mm capstan give about 6.7 t with the thrust ball bearing (3.6 t with a plain collar) | Physics | Mechanics and statics | Power screw torque T = F (dm / 2) tan(lambda + phi) plus collar friction; the detailed model's thrust ball bearing cuts the collar term, giving about 6.7 t | claimed 6.7 t, estimate 6.7 t (ratio 1.00) | **Within known science** | PPR-OPT-001 v0.3 (thrust bearing in the detailed model) |
| OC2 | Option C: the Tr40 x 7 screw self-locks (lead angle below the friction angle) | Physics | Mechanics and statics | Self-locking when lambda < phi | 3.49333 deg <= 8.827 deg | **Within known science** | This file |
| OC3 | Option C: about 43 | Mathematics | Orders of magnitude | Turns = travel / lead | claimed 43 turns, estimate | | PPR-OPT-001 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | turns cover the 300 mm opening | | | | 42.9 turns (ratio 1.00) | **Within known science** | |
| OC4 | Option C: the screw does not buckle at 1.5 times 10 t over its free length of about 0.4 m closed (0.7 m open) | Physics | Strength of materials | Euler buckling, pinned at the nut and the thrust swivel, P = pi^2 E I / L^2 on the root diameter, at the open length as the worst case | 147.1 kN <= 207.3 kN | **Within known science** | This file (model: 330 mm swivel to crown closed, plus the 310 mm opening) |

## The four fixes (PPR-DDR-006)

1. **Clay shrinkage (M6, D2).** The reference filter (280 mm inner rim, 240 mm deep, 15 mm wall) is now the fired size, and the mold cavities are that size divided by (1 - 0.08) for an assumed 8 % total linear shrinkage. The cavities are 304 mm at the rim and 261 mm deep, with a 16.3 mm wall. The 8 % is the middle of the 6 to 10 % that earthenware filter mixes show. It is one parameter ( `CLAY_SHRINK` in `cad/src/model.py` ), to be replaced by a test bar on the partner's production mix fired in its own kiln (ASTM C326 method) before any pattern is printed. A 2 % error moves the fired rim by about 6 mm. Knock-on changes: mold flanges 450 to 480 mm, flash groove radius 195 mm, dowels at 216 mm, frame 22 mm taller (overall 1,828 mm). The charge is now 9.6 kg, against the 8 to 9.5 kg factories report, and the fired pot is 4.9 kg. The QC rack holds fired pots and is unchanged.
2. **Heavy lifts (B2, D3).** Every part over 25 kg goes on a 0.5 t chain block (BOM item 22). The female mold lifts by two M12 DIN 580 eye bolts in its steel base plate, and the male mold by two in its flange top, each on a two-leg sling with shackles. The beams and platen go in a round sling. Eye bolts come out and the holes are plugged before pressing.
3. **Pump handle force (S8, D4).** The working force was limited to 6.4 t, where the frictionless handle force reaches the 150 N screening limit (PR-004). With 85 % pump and seal efficiency counted, the limit is 5.4 t (D9, decided by Amish on 2026-10-25). If pressing trials need more, a separate two-speed hand pump at waist height goes to Amish as decision D6.
4. **Scrap alloy (C3, D5).** The molds use A356 cast alloy wheel scrap only, sorted before melting to remove lead balance weights, brass valve stems, steel and paint. A chill sample of each melt is checked by spark OES, or by handheld XRF for lead, before pouring. Lead and cadmium must be within the EN 601 food-contact limits, recorded per heat (PR-012). The limit values are still to confirm.

## Corrections made to earlier numbers

- **Fired pot mass (C1).** Rice husk ash (about 18 % of the husk) is now in PPR-CAL-001; with the larger pressed pot the fired pot is 4.9 kg.
- **Cast aluminium yield.** Lowered from 90 to 80 MPa, as-cast remelted A356 without heat treatment. The male flange works at 60 MPa at 1.5 times the jack rating.
- **Joint bolts (S3).** Checked against the bolt shear resistance of 400 MPa (EN 1993-1-8).

- **Tipping (M5).** Restoring moment 850 N m with the heavier molds and the pressed pot taken at 8.72 kg (independent review, 2026-10-24).
- **Printer (E1).** Each pattern quadrant is now 243 mm on a 250 mm bed; with the 2 degree flange draft about 5 mm is left, so a true 250 x 250 mm bed is required.
- **Guards (E3).** ISO 13857 Table 7 (lower limbs) and the floor gap are still outside the desk check.

## Independent review of the fixes

A second agent, given the model, the calculations, the STEM script and the BOM but not the reasoning, tried to refute each fix (2026-10-23).

- **Shrinkage:** held. Uniform scaling is right to first order, the casting shrink still applies on top, travel (91 of 110 mm) and opening (291 of 310 mm) still work. It noted that the larger pot needs about 11.8 t for the mean pressure of the earlier design (now 0.57 MPa at the 6.4 t limit), that the charge is just above the factory range, and that the BOM text was stale. All three are now recorded or fixed.
- **Lifting:** weak as first drawn. The female eye holes sat in the lapped stop face, a fitted eye bolt would be crushed under the male flange, and the eye collar overhung the flange edge. Corrected: female lift points moved to the steel base plate, male holes to 214 mm, a two-leg sling with shackles, and holes plugged before pressing.
- **Handle force:** refuted as first drawn, with a 400 mm slip-on extension. A sleeve must overlap the handle, the extended tip reached about 61 mm off the floor and crossed the QC rack, and jack makers prohibit extensions. Corrected: no extension; the working force is limited to 6.4 t, and a remote two-speed pump is proposed to Amish (D6).
- **Scrap alloy:** weak. A melt test alone catches contamination too late, XRF misses magnesium and reads silicon poorly, and LM6 has little strength margin. Corrected: scrap sorted first, OES preferred, A356 only, yield taken as 80 MPa.

## Independent review, 2026-10-24 (kit 1.9.1 upgrade)

A second agent, given only this check, the calculation note PPR-CAL-001, the product requirements PPR-PRS-001 and `docs/04-calcs/sizing.py`, and none of the author's reasoning, tried to refute the headline results and every "Within known science" verdict, including the new option rows. It reproduced S1, S3, S4, S6, M1 to M6, C1, C5, OB1, OB2, OC1 to OC3 and the tipping moment, and raised nine findings. What changed:

| # | Finding | Action |
|---|---|---|
| 1 | Working force stated as 10 t in PR-003, S3, S4, S6 and S9 but capped at 6.4 t in PR-004 and S8; at 6.4 t the clay sees 0.57 MPa, and the earlier design's 1.05 MPa would need about 11.8 t | S5 restated: 2 t forms a filter on a low-cost press, inside the 6.4 t limit, and the force this larger pot needs is found in trials. Stating 10 t as a structural check case and 6.4 t as the only working force means changing PR-003 text Amish decided on 2026-09-25, so it went to Amish as **D9**; decided by Amish on 2026-10-25 as recommended (PR-003 restated, PR-004 at 5.4 t) |
| 2 | S8 has no margin: 149 N against 150 N frictionless, about 175 N with 85 % pump and seal efficiency, about 50 strokes a minute, no EN 1005-3 multipliers | **Accepted.** S8 is now Plausible, to confirm by an EN 1005-3 assessment with the bought jack's data. The remedy (working force limit 5.4 t, or the two-speed pump of D6 as the baseline) changes what the press offers the factory, so it went to Amish in **D9**; Amish decided the 5.4 t limit on 2026-10-25 (148 N with efficiency) |
| 3 | S7 power not traceable | It is computed in `stem_check.py` (load building over the last 20 mm at 10 t, 1 min of pumping); now with 85 % efficiency, 28 W against 75 W |
| 4 | Deflection omits bolt slip in 18 mm holes and pin play | **Fixed under the fix rule:** the build plan now tightens the M16 10.9 joint bolts to preload (about 300 N·m, 110 kN); new row S11 shows 36.8 kN per bolt against 44 kN of slip resistance, so the joints do not slip at 294 kN. Pin play is taken up before load and the molds close on a metal stop, so it does not set the wall |
| 5 | Frame cases not checked: moment at the bolted joints, channel torsion from the web-plane pin load, net section at the bolt holes, the doubler welds, a stress concentration of about 1.5 to 2 at the male mold's plug root (60 MPa nominal against 80 MPa at 1.5 times the jack rating) | Not resolved this session. They need a frame and mold analysis beyond first order (a finite element check of the male mold root and the top beam joint); proposed to Amish as the next calculation step (REVIEW.md). At the 6.4 t working force the plug root works at about 13 MPa nominal, about 26 MPa with the concentration (less at the 5.4 t limit decided on 2026-10-25) |
| 6 | Option B pins checked in shear only; about 510 MPa in bending at 30 mm; the force is not bounded by the operator | **Fixed in the option B model:** pins 30 to 40 mm, links 80 mm wide, bushes 50 x 40 (OB3, OB4: 199 MPa in bending against 512 MPa). The overcharge limit is a risk of option B in PPR-OPT-001 |
| 7 | Option C screw buckling and thrust bearing capacity unchecked | Row OC4: 207 kN Euler load on the root diameter at the open free length of about 0.7 m against 147 kN. The thrust bearing's static rating at the two-person force is "to assess" in PPR-OPT-001 |
| 8 | Stale numbers: a 7.4 kg charge in the tipping sum, 39.7 against 39.6 kg, the PRS introduction at $1,169, 7.5 kg of mix and CAL v0.10 | **Fixed:** the moving mass takes the 8.72 kg pressed pot (tipping push 850 N, demold lugs 11.8 kN, PPR-CAL-001 v0.12); 39.6 kg throughout; PRS v0.15 brought up to date. The melt-check fee is not yet in the BOM (noted in REVIEW.md) |
| 9 | The 54 kg demold tilt is manual and outside the chain-block rule | Not changed: the tipping pins and lift handles were decided by Amish on 2026-10-02, and the start of the tip needs about 308 N on the two handles. Recorded in REVIEW.md as a handling risk for the partner trial |

The reviewer's challenge to S2 (a longer pin bending arm from the 28 mm screw hole) was checked and not accepted: the hole takes the load off the middle of the pin block and moves it toward the webs, which shortens the arm, so 272 MPa stays conservative.

## Fields not relevant

None. Every field applies: physics (structure, hydraulics, statics), chemistry (clay, firing, alloys, silver), biology (ergonomics, dust and fume, the filter's microbial role), mathematics (geometry, rates, balances) and engineering (printing, finishing, guarding, lifting).

## Concept options

- **Option A, hydraulic bottle-jack press:** this check.
- **Option B, toggle lever press:** within known science (rows OB1 to OB3, checked on the detailed model). With the 1.1 m lever rigid on the upper link and 300 N at its grip, the toggle gives F = T / (2 L sin theta): about 0.16 t at 30 mm from straight, 1.4 t at 0.35 mm and 7 t at 0.014 mm. The force at the stop is therefore set by the clay charge, not by the operator (PPR-OPT-001). The 70 x 40 mm lower link and the 30 mm clevis pins carry 1.5 times 10 t with a wide margin.
- **Option C, power-screw press:** within known science (rows OC1 to OC3). A Tr40 x 7 screw with a thread friction coefficient of 0.15 and the detailed model's thrust ball bearing gives about 6.7 t for two people pulling 300 N at a 450 mm capstan radius (about 3.6 t with a plain collar). The screw self-locks (lead angle 3.5 degrees against a friction angle of 8.8 degrees); about 43 turns cover a 300 mm opening (PPR-OPT-001).

## Sources


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